

Question 1:

Solve the following system of linear equations using LU decomposition:

$$\begin{aligned}3x + y - 2z &= 9 \\2x + 4y + z &= 8 \\-x + 3y + 3z &= 4\end{aligned}$$

Question 2:

Given the matrix A and vector b , perform LU decomposition and solve for x :

$$A = \begin{bmatrix} 5 & 2 & 3 \\ 3 & 8 & 1 \\ 4 & 1 & 6 \end{bmatrix}, \quad b = \begin{bmatrix} 7 \\ 15 \\ 10 \end{bmatrix}$$

Question 3:

Solve the system of equations:

$$4x + 3y - z = 2$$

$$2x + y + 3z = 3$$

$$5x - 2y + 4z = 6$$

using LU decomposition.

Question 4:

Solve the system of equations using LU decomposition:

$$2x + 3y + 4z = 10$$

$$3x + 2y + z = 8$$

$$x + 4y + 2z = 7$$

Question 5:

Given the LU decomposition:

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 4 & 5 & 1 \end{bmatrix}, \quad U = \begin{bmatrix} 4 & -3 & 2 \\ 0 & 5 & 1 \\ 0 & 0 & 3 \end{bmatrix}, \quad b = \begin{bmatrix} 8 \\ 10 \\ 6 \end{bmatrix}$$

Solve for x using LU decomposition.

Question 6:

Given the following system of linear equations, perform LU decomposition and then solve the system:

$$\begin{aligned}2x + y + z &= 5 \\4x + 3y - 2z &= 6 \\6x - y + 3z &= 7\end{aligned}$$

Question 7:

Solve the following system using LU decomposition:

$$\begin{aligned}x + 2y + 3z &= 10 \\2x + y + z &= 8 \\3x - y + 2z &= 7\end{aligned}$$

Question 8:

For the following system, perform LU decomposition and solve for x :

$$2x + 4y + 5z = 10$$

$$3x + 5y + 4z = 8$$

$$4x + 2y + 3z = 7$$

Question 9:

Given the following system of equations, use LU decomposition to solve for x , y , and z :

$$3x + 2y + 5z = 15$$

$$2x + 4y + 3z = 10$$

$$5x + y + 2z = 12$$

Question 10:

Solve the following system of equations using LU decomposition:

$$x - y + 2z = 4$$

$$2x + y - z = 2$$

$$3x + 2y + z = 7$$